

A Gibbs–Greedy $N^{2/3}$ Construction for Sine Sums and Riesz-Product Consequences of Two-Dissociation

Xiyu Hu

August 2, 2026

Abstract

For a finite set $A \subset \mathbb{N}$, write

$$S_A(x) = \sum_{a \in A} \sin(ax), \quad x \in \mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z}).$$

This note has three purposes. First, it gives a self-contained Gibbs-potential proof that for every N there is a set A of N distinct positive integers such that

$$\|S_A\|_\infty \ll N^{2/3}, \quad \max A \leq \exp(O(N^{1/3})).$$

The exponent $2/3$ is the known Bourgain order; the proof presented here is a deterministic logarithmic-window selection argument. Second, two rigorous consequences of finite Riesz products and two-dissociation are proved:

$$\|S_A\|_\infty \geq \frac{\dim_2(A)^2}{4|A|},$$

and, for a probability density f on the circle,

$$\dim_2(\text{Spec}_\rho(f)) \leq \frac{4 \text{Ent}(f)}{\rho^2}.$$

The latter applies directly to the Gibbs measures generated during the construction. Third, affine antiphase blocks, relative two-dissociation, and orbit-reflection blocks are developed as a conditional route beyond $2/3$. Exact factorisations and conditional block estimates are established, while the unresolved points are stated explicitly. Numerical experiments and reproducible Python code accompany the note.

Status and scope. The $N^{2/3}$ theorem below reproduces a known order of magnitude by a self-contained argument; it is not presented as a new bound. The two Riesz-product inequalities are elementary deductions proved here in full. The final affine-block section is exploratory and conditional: no unconditional improvement over $2/3$ is claimed.

Contents

1	The sine-sum problem	2
2	A closed Gibbs–greedy construction at the $N^{2/3}$ scale	2
2.1	A uniform harmonic sine estimate	3
2.2	One-step logarithmic-window selection	3
2.3	Iteration and conversion from soft maximum to maximum	4

3	Riesz products and two-dissociation	5
3.1	Fourier uniqueness and a clean tail calculation	5
3.2	First substantive consequence: two-dissociative dimension obstructs flatness	7
3.3	Second substantive consequence: entropy controls the two-dimensional large spectrum	8
4	Affine antiphase blocks and a conditional route beyond $2/3$	9
4.1	Simple affine antiphase pairs	9
4.2	What a uniform block lemma would imply	11
4.3	Orbit-reflection blocks and second-order vanishing	11
4.4	The remaining obstacles	12
5	Numerical experiments	13
5.1	Bessel–Gibbs greedy search	13
5.2	Gibbs-spectrum diagnostics	13
5.3	A sample affine-block diagnostic	14
6	Conclusions	14

1 The sine-sum problem

Throughout, m denotes normalized Haar measure on $\mathbb{T}$:

$$dm(x) = \frac{dx}{2\pi}.$$

For a finite $A \subset \mathbb{N}$, define

$$S_A(x) = \sum_{a \in A} \sin(ax).$$

Since S_A is odd,

$$\|S_A\|_\infty = \max_{x \in \mathbb{T}} S_A(x) = -\min_{x \in \mathbb{T}} S_A(x).$$

The classical extremal question asks for the size of

$$M_{\sin}(N) := \inf_{\substack{A \subset \mathbb{N} \\ |A|=N}} \|S_A\|_\infty.$$

Orthogonality gives the elementary lower bound $M_{\sin}(N) \geq \sqrt{N/2}$. Bourgain proved that $M_{\sin}(N) \ll N^{2/3}$; this is recorded, for example, by Kolountzakis [1]. The next section gives a complete deterministic proof of the same exponent.

2 A closed Gibbs–greedy construction at the $N^{2/3}$ scale

The construction uses two ideas:

- (i) a Gibbs integral records all dangerous high points simultaneously;
- (ii) a logarithmic average over candidate frequencies is controlled by the uniformly bounded harmonic sine sums.

2.1 A uniform harmonic sine estimate

Lemma 2.1 (Uniform boundedness of harmonic sine blocks). *There is an absolute constant C_0 such that for all integers $1 \leq u \leq v$ and every $x \in \mathbb{T}$,*

$$\left| \sum_{n=u}^v \frac{\sin(nx)}{n} \right| \leq C_0.$$

Proof. Let

$$\delta = \text{dist}(x, 2\pi\mathbb{Z}) \in [0, \pi].$$

The case $\delta = 0$ is immediate. For $n \leq \delta^{-1}$,

$$|\sin(nx)| \leq n\delta,$$

so the contribution from this range is $O(1)$. For an interval $[a, b]$ with $a > \delta^{-1}$, the geometric-series estimate gives

$$\sup_{a \leq r \leq b} \left| \sum_{n=a}^r \sin(nx) \right| \ll \delta^{-1}.$$

Abel summation therefore yields

$$\left| \sum_{n=a}^b \frac{\sin(nx)}{n} \right| \ll \frac{1}{a\delta} + \frac{1}{\delta} \sum_{n=a}^{b-1} \left(\frac{1}{n} - \frac{1}{n+1} \right) \ll \frac{1}{a\delta} \ll 1.$$

Combining the two ranges proves the claim. $\square$

2.2 One-step logarithmic-window selection

Fix an intended final cardinality N and a parameter $0 < \beta \leq 1$. Put

$$L = 2N + 1, \quad M = \lceil L \exp(K/\beta) \rceil,$$

where K is a sufficiently large absolute constant.

Lemma 2.2 (One-step Gibbs selection). *Let μ be any probability measure on $\mathbb{T}$, and let $E \subset \mathbb{N}$ have $|E| \leq N$. Then there exists*

$$n \in [L, M] \cap \mathbb{N} \setminus E$$

such that

$$\int_{\mathbb{T}} e^{\beta \sin(nx)} d\mu(x) \leq \exp(C\beta^2),$$

where C is absolute.

Proof. Let

$$\mathcal{I} = [L, M] \cap \mathbb{N} \setminus E, \quad H = \sum_{n \in \mathcal{I}} \frac{1}{n}.$$

Since

$$\sum_{n=L}^M \frac{1}{n} \geq \int_L^{M+1} \frac{dt}{t} \geq \frac{K}{\beta},$$

and the deleted harmonic mass is at most

$$\sum_{n \in E \cap [L, M]} \frac{1}{n} \leq \frac{N}{2N+1} < \frac{1}{2},$$

we have, after increasing K if necessary,

$$H \geq \frac{K}{2\beta}.$$

By [theorem 2.1](#), for every x ,

$$\left| \sum_{n \in \mathcal{I}} \frac{\sin(nx)}{n} \right| \leq C_0 + \frac{1}{2}.$$

Consequently,

$$\frac{1}{H} \sum_{n \in \mathcal{I}} \frac{\sin(nx)}{n} \leq C_1 \beta.$$

For $|u| \leq 1$ and $0 < \beta \leq 1$,

$$e^{\beta u} \leq 1 + \beta u + C_2 \beta^2.$$

Hence, pointwise in x ,

$$\frac{1}{H} \sum_{n \in \mathcal{I}} \frac{1}{n} e^{\beta \sin(nx)} \leq 1 + C_3 \beta^2.$$

Integrating against μ and interchanging the finite sum and integral gives

$$\frac{1}{H} \sum_{n \in \mathcal{I}} \frac{1}{n} \int e^{\beta \sin(nx)} d\mu(x) \leq 1 + C_3 \beta^2 \leq e^{C_4 \beta^2}.$$

The left-hand side is a positive weighted average, so at least one candidate n satisfies the desired estimate. $\square$

2.3 Iteration and conversion from soft maximum to maximum

Theorem 2.3 (A deterministic $N^{2/3}$ construction). *For every $N \geq 1$, there is a set $A \subset \mathbb{N}$ with $|A| = N$ such that*

$$\|S_A\|_\infty \leq CN^{2/3}.$$

Moreover, A can be chosen inside

$$\left[1, (2N+1)e^{CN^{1/3}} + 1 \right].$$

Proof. Fix $0 < \beta \leq 1$, to be chosen later. Starting from $A_0 = \emptyset$, suppose that distinct frequencies $A_j = \{n_1, \dots, n_j\}$ have been chosen. Define

$$F_j(x) = \sum_{r=1}^j \sin(n_r x), \quad Z_j = \int e^{\beta F_j(x)} dm(x),$$

and the Gibbs probability measure

$$d\mu_j(x) = \frac{e^{\beta F_j(x)}}{Z_j} dm(x).$$

Apply [theorem 2.2](#) with $E = A_j$ to choose a new frequency $n_{j+1} \notin A_j$. Then

$$\begin{aligned} Z_{j+1} &= \int e^{\beta F_j(x)} e^{\beta \sin(n_{j+1}x)} \, dm(x) \\ &= Z_j \int e^{\beta \sin(n_{j+1}x)} \, d\mu_j(x) \\ &\leq Z_j e^{C\beta^2}. \end{aligned}$$

Since $Z_0 = 1$, after N steps,

$$\log Z_N \leq CN\beta^2.$$

All selected frequencies lie in $[L, M]$, so their maximum D satisfies

$$\log D \leq C\beta^{-1} + C \log N.$$

Let

$$Q = F_N, \quad \mathcal{M} = \|Q\|_\infty = \max Q.$$

The degree of Q is at most D . Bernstein's inequality gives

$$\|Q'\|_\infty \leq D \|Q\|_\infty = D\mathcal{M}.$$

If $Q(x_0) = \mathcal{M}$, then

$$Q(x) \geq \frac{\mathcal{M}}{2} \quad \text{whenever} \quad |x - x_0| \leq \frac{1}{2D}.$$

Thus

$$Z_N \geq \frac{c}{D} e^{\beta \mathcal{M}/2},$$

and hence

$$\mathcal{M} \leq \frac{C}{\beta} (\log Z_N + \log D + 1) \leq C (N\beta + \beta^{-2} + \beta^{-1} \log N).$$

Choose $\beta = N^{-1/3}$ for N sufficiently large. Then

$$N\beta = \beta^{-2} = N^{2/3}, \quad \beta^{-1} \log N = N^{1/3} \log N \ll N^{2/3}.$$

The finitely many small N are absorbed into the constant. □

Remark 2.4 (The origin of the exponent). The proof balances two costs:

$$\underbrace{N\beta}_{\text{accumulated Gibbs-potential growth}} \quad \text{and} \quad \underbrace{\beta^{-2}}_{\text{frequency-range / peak-width cost}}.$$

The balance $N\beta \asymp \beta^{-2}$ forces $\beta \asymp N^{-1/3}$ and gives $N^{2/3}$.

3 Riesz products and two-dissociation

3.1 Fourier uniqueness and a clean tail calculation

Definition 3.1 (Two-dissociation). A finite set $D \subset \mathbb{Z}$ is *two-dissociated* if

$$\sum_{d \in D} c_d d = 0, \quad c_d \in \{-2, -1, 0, 1, 2\},$$

implies $c_d = 0$ for every $d \in D$. For $V \subset \mathbb{Z}$, let $\dim_2(V)$ be the maximum cardinality of a two-dissociated subset of V . This is the convention used by Bedert [2].

Let D be two-dissociated, $|D| = d$, and let

$$R_{p,\phi}(x) = \prod_{m \in D} (1 + p \cos(mx + \phi_m)), \quad 0 \leq p \leq 1.$$

Lemma 3.2 (Unique Fourier words). *The map*

$$\varepsilon \in \{-1, 0, 1\}^D \mapsto \sum_{m \in D} \varepsilon_m m$$

is injective. Consequently, distinct terms in the full Fourier expansion of $R_{p,\phi}$ have distinct frequencies, and

$$\int R_{p,\phi} \, dm = 1.$$

Proof. If two words $\varepsilon, \eta \in \{-1, 0, 1\}^D$ produce the same frequency, then

$$\sum_{m \in D} (\varepsilon_m - \eta_m) m = 0,$$

with coefficients in $\{-2, -1, 0, 1, 2\}$. Two-dissociation implies $\varepsilon = \eta$. The zero frequency therefore arises only from the all-zero word. $\square$

Write

$$R_{p,\phi} = 1 + R_1 + R_{\geq 2}, \quad R_1 = p \sum_{m \in D} \cos(mx + \phi_m).$$

Proposition 3.3 (Exact L^2 size of the Riesz tail). *One has*

$$\|R_{\geq 2}\|_2^2 = \left(1 + \frac{p^2}{2}\right)^d - 1 - \frac{dp^2}{2}.$$

In particular, if $dp^2 \leq 1$, then

$$\|R_{\geq 2}\|_2 \ll dp^2.$$

Proof. A word with support size k has Fourier coefficient of magnitude $(p/2)^k$. There are $\binom{d}{k} 2^k$ such words, and by [theorem 3.2](#) all their frequencies are distinct. Parseval therefore gives

$$\|R_{\geq 2}\|_2^2 = \sum_{k=2}^d \binom{d}{k} 2^k \left(\frac{p}{2}\right)^{2k} = \sum_{k=2}^d \binom{d}{k} \left(\frac{p^2}{2}\right)^k.$$

This is the stated binomial expression. The final estimate follows from the Taylor expansion of $(1+t)^d$ when $dt \leq 1$. $\square$

Corollary 3.4 (Aligned first-order term with controlled tail). *Let $f \in L^2(m)$ satisfy $f \geq 0$ and $\int f \, dm = 1$. Phases ϕ_m can be chosen so that*

$$\int f(x) \cos(mx + \phi_m) \, dm(x) = -|\widehat{f}(m)|.$$

For these phases,

$$\int f R_{p,\phi} \, dm \leq 1 - p \sum_{m \in D} |\widehat{f}(m)| + \|f\|_2 \left[\left(1 + \frac{p^2}{2}\right)^d - 1 - \frac{dp^2}{2} \right]^{1/2}.$$

Proof. Choose the phases to align the first-order Fourier coefficients negatively. Then use

$$\int f R_{p,\phi} = 1 - p \sum_{m \in D} |\widehat{f}(m)| + \int f R_{\geq 2}$$

and apply Cauchy–Schwarz together with [theorem 3.3](#). $\square$

3.2 First substantive consequence: two-dissociative dimension obstructs flatness

Theorem 3.5 (A two-dissociative obstruction). *Let $A \subset \mathbb{N}$ be finite, $|A| = N$, and let $d = \dim_2(A)$. Then*

$$\|S_A\|_\infty \geq \frac{d^2}{4N}.$$

Equivalently, any family satisfying $\|S_A\|_\infty \leq M$ must satisfy

$$\dim_2(A) \leq 2\sqrt{MN}.$$

Proof. Choose a two-dissociated subset $D \subset A$ with $|D| = d$, and define

$$R_p(x) = \prod_{m \in D} (1 - p \sin(mx)), \quad 0 \leq p \leq 1.$$

Each factor is nonnegative, and [theorem 3.2](#) implies

$$\int R_p \, dm = 1.$$

Thus $R_p \, dm$ is a probability measure.

For $m \in D$, Fourier-word uniqueness shows that the only contribution to frequency $\pm m$ is the linear term $-p \sin(mx)$. Therefore

$$\int \sin(mx) R_p(x) \, dm(x) = -\frac{p}{2}.$$

Now let $a \in A \setminus D$. If the Fourier coefficient of R_p at a is nonzero, then a has a representation

$$a = \sum_{m \in D} \varepsilon_m m, \quad \varepsilon_m \in \{-1, 0, 1\}.$$

The support of this representation has size at least two, since $a \notin D$. By two-dissociation the representation is unique, so

$$\left| \int \sin(ax) R_p(x) \, dm(x) \right| \leq \frac{p^2}{4}.$$

Consequently,

$$\int S_A(x) R_p(x) \, dm(x) \leq -\frac{pd}{2} + \frac{(N-d)p^2}{4}.$$

Take $p = d/N$. Then

$$\int S_A R_p \, dm \leq -\frac{d^2}{4N}.$$

Since $R_p \, dm$ is a probability measure, some x satisfies

$$S_A(x) \leq -\frac{d^2}{4N}.$$

Oddness of S_A completes the proof. $\square$

Remark 3.6. This theorem shows that making the final frequency set highly two-dissociated is incompatible with flatness. Two-dissociation is useful as a device for controlling Fourier words and tails, but a fully two-dissociated final set has a large, in fact linear-scale, sine peak.

3.3 Second substantive consequence: entropy controls the two-dimensional large spectrum

For a probability density f with respect to m , define

$$\text{Ent}(f) = \int_{\mathbb{T}} f \log f \, dm$$

with the usual convention $0 \log 0 = 0$. Define the large spectrum

$$\text{Spec}_\rho(f) = \{n \in \mathbb{Z} \setminus \{0\} : |\widehat{f}(n)| \geq \rho\}, \quad 0 < \rho \leq 1.$$

Theorem 3.7 (Riesz-product entropy bound). *For every probability density f and every $0 < \rho \leq 1$,*

$$\dim_2(\text{Spec}_\rho(f)) \leq \frac{4 \text{Ent}(f)}{\rho^2}.$$

If $\text{Ent}(f) = +\infty$, the assertion is understood as trivial.

Proof. Let $D \subset \text{Spec}_\rho(f)$ be two-dissociated, with $|D| = d$. Choose phases ϕ_n such that

$$\int f(x) \cos(nx + \phi_n) \, dm(x) = |\widehat{f}(n)| \geq \rho.$$

For $0 \leq p \leq 1/2$, set

$$R(x) = \prod_{n \in D} (1 + p \cos(nx + \phi_n)).$$

By [theorem 3.2](#), $R \, dm$ is a probability measure. Nonnegativity of relative entropy gives

$$0 \leq \int f \log \frac{f}{R} \, dm = \text{Ent}(f) - \int f \log R \, dm.$$

For $|u| \leq 1$ and $0 \leq p \leq 1/2$,

$$\log(1 + pu) \geq pu - p^2.$$

Therefore

$$\begin{aligned} \text{Ent}(f) &\geq \sum_{n \in D} \int f(x) \log(1 + p \cos(nx + \phi_n)) \, dm(x) \\ &\geq d(p\rho - p^2). \end{aligned}$$

Taking $p = \rho/2$ gives

$$\text{Ent}(f) \geq \frac{d\rho^2}{4}.$$

Maximizing over two-dissociated D proves the theorem. □

Corollary 3.8 (Large spectrum of a Gibbs state). *Let*

$$F(x) = \sum_{j=1}^J \sin(n_j x), \quad f_\beta(x) = \frac{e^{\beta F(x)}}{\int e^{\beta F} dm}, \quad 0 < \beta \leq 1.$$

Then

$$\dim_2(\text{Spec}_\rho(f_\beta)) \leq \frac{4\beta J}{\rho^2}.$$

Proof. Writing $Z = \int e^{\beta F} dm$, one has

$$\text{Ent}(f_\beta) = \beta \int F f_\beta dm - \log Z.$$

Since $F \leq J$ and Jensen gives $\log Z \geq \beta \int F dm = 0$,

$$\text{Ent}(f_\beta) \leq \beta J.$$

Apply [theorem 3.7](#). □

4 Affine antiphase blocks and a conditional route beyond 2/3

The results in this section separate rigorous identities and conditional estimates from the still-unproved selection principle needed for an improved exponent.

4.1 Simple affine antiphase pairs

Let q be a positive even integer and let $D = \{d_1, \dots, d_r\} \subset \{1, \dots, q-1\}$. Define

$$G_{q,D}(x) = \sum_{d \in D} (\sin(dx) + \sin((q-d)x)).$$

The elementary factorisation

$$\boxed{G_{q,D}(x) = 2 \sin \frac{qx}{2} \sum_{d \in D} \cos \left(d - \frac{q}{2} \right) x} \tag{4.1}$$

shows that the block is small where qx is close to a multiple of 2π .

Let μ be a probability measure and put

$$\delta_q(\mu) = 1 - \Re \hat{\mu}(q).$$

If $\delta_q(\mu) > 0$, define the defect measure

$$d\nu_q(x) = \frac{2 \sin^2(qx/2)}{\delta_q(\mu)} d\mu(x).$$

This is a probability measure.

Proposition 4.1 (Relative second-moment estimate). *Put $a_i = d_i - q/2$. Suppose that, for some $0 \leq \rho \leq 1$,*

$$\begin{aligned} |\widehat{\nu}_q(2a_i)| &\leq \rho, \\ |\widehat{\nu}_q(a_i - a_j)| &\leq \rho, \quad |\widehat{\nu}_q(a_i + a_j)| \leq \rho \quad (i \neq j). \end{aligned}$$

Then

$$\int G_{q,D}^2 \, d\mu \leq \delta_q(\mu)r + 2\delta_q(\mu)\rho r^2.$$

In particular, if $\rho \leq r^{-1}$ and $\delta_q(\mu) \leq r^{-\eta}$, then

$$\int G_{q,D}^2 \, d\mu \ll r^{1-\eta}.$$

Proof. Let

$$H(x) = \sum_{i=1}^r \cos(a_i x).$$

By (4.1),

$$\int G_{q,D}^2 \, d\mu = 2\delta_q(\mu) \int H^2 \, d\nu_q.$$

Using

$$\cos^2 u = \frac{1 + \cos(2u)}{2}, \quad 2 \cos u \cos v = \cos(u - v) + \cos(u + v),$$

the stated Fourier assumptions give

$$\int H^2 \, d\nu_q \leq \frac{r}{2} + \rho r^2.$$

Multiplication by $2\delta_q(\mu)$ proves the result. $\square$

The role envisioned for two-dissociation is now precise: it should help select D so that the frequencies

$$2a_i, \quad a_i - a_j, \quad a_i + a_j$$

avoid the large spectrum of the defect measure ν_q . Two-dissociation alone does not imply this relative Fourier avoidance; it supplies uniqueness of low-order words once the relevant large-spectrum obstructions have been removed.

Proposition 4.2 (Conditional Gibbs gain from an affine block). *Assume the hypotheses of [theorem 4.1](#), and suppose additionally that*

$$\int G_{q,D} \, d\mu \leq 0, \quad \beta r \leq c_0$$

for a sufficiently small absolute constant c_0 . Then

$$\log \int e^{\beta G_{q,D}} \, d\mu \leq C\beta^2 \delta_q(\mu)(r + \rho r^2).$$

Proof. Since $|G_{q,D}| \leq 2r$ and $\beta r \leq c_0$,

$$e^{\beta G_{q,D}} \leq 1 + \beta G_{q,D} + C\beta^2 G_{q,D}^2.$$

The linear term is nonpositive. Apply [theorem 4.1](#) and use $\log(1+t) \leq t$. $\square$

4.2 What a uniform block lemma would imply

Proposition 4.3 (Conditional exponent improvement). *Fix $\eta > 0$. Suppose that at every Gibbs stage, one can select a fresh block of $2r$ frequencies, with*

$$r \asymp \beta^{-1},$$

whose block polynomial G satisfies

$$\int G \, d\mu \leq 0, \quad \int G^2 \, d\mu \ll r^{1-\eta}, \quad \|G\|_\infty \leq 2r.$$

Assume also that all selected frequencies remain below $\exp(C/\beta)$. Then one obtains sets A , $|A| = N$, satisfying

$$\|S_A\|_\infty \ll N^{2/(3+\eta)}.$$

Proof. For βr sufficiently small, each block increases $\log Z$ by at most

$$C\beta^2 r^{1-\eta}.$$

There are $O(N/r)$ blocks, so

$$\log Z_N \ll N\beta^2 r^{-\eta} \asymp N\beta^{2+\eta}.$$

The frequency bound and Bernstein's inequality give

$$\|S_A\|_\infty \ll \frac{\log Z_N + \beta^{-1}}{\beta} \ll N\beta^{1+\eta} + \beta^{-2}.$$

Balancing the two terms gives

$$\beta \asymp N^{-1/(3+\eta)}$$

and hence the claimed exponent. □

Thus any fixed power gain in the effective block variance would break the $2/3$ barrier. The unproved part is a uniform selection theorem producing such blocks for every Gibbs state encountered in the iteration.

4.3 Orbit-reflection blocks and second-order vanishing

A stronger affine construction is obtained by completing an arithmetic orbit before reflecting it. Let

$$q = Lh, \quad h \in 2\mathbb{N},$$

and choose

$$D \subset \{1, \dots, h/2 - 1\}, \quad |D| = r.$$

Define the frequency block

$$\mathcal{B}(q, L, D) = \bigcup_{\substack{d \in D \\ 0 \leq j < L}} \{d + jh, \, q - d - jh\}.$$

All these frequencies are positive and distinct, and $|\mathcal{B}| = 2Lr$. Put

$$G_{q,L,D}(x) = \sum_{n \in \mathcal{B}(q,L,D)} \sin(nx).$$

Proposition 4.4 (Exact orbit-reflection factorisation). *One has*

$$G_{q,L,D}(x) = 2 \frac{\sin^2(qx/2)}{\sin(hx/2)} \sum_{d \in D} \cos\left(d - \frac{h}{2}\right)x \quad (4.2)$$

with continuous extension at the zeros of $\sin(hx/2)$.

Proof. For each $d \in D$,

$$\sum_{j=0}^{L-1} e^{i(d+jh)x} = e^{idx} \frac{1 - e^{iqx}}{1 - e^{ihx}},$$

whereas

$$\sum_{j=0}^{L-1} e^{i(q-d-jh)x} = e^{i(h-d)x} \frac{1 - e^{iqx}}{1 - e^{ihx}}.$$

Adding, rewriting the two elementary factors in sine form, and taking imaginary parts gives (4.2). $\square$

If

$$qx = 2\pi k + t, \quad t \rightarrow 0,$$

then for $L \nmid k$ the denominator in (4.2) tends to $\sin(\pi k/L) \neq 0$, while the numerator is $O(t^2)$. Hence

$$G_{q,L,D}(x) = O(t^2) \quad (L \nmid k).$$

On the one coarse fibre $L \mid k$, the denominator also vanishes and only first-order cancellation remains. The block therefore has second-order vanishing on $L - 1$ of the L residue fibres.

There is, however, an unavoidable escape peak. At

$$x_* = \frac{\pi}{2q},$$

each pair $\{n, q - n\}$ contributes

$$\sin(nx_*) + \sin((q - n)x_*) \geq 1.$$

Therefore

$$G_{q,L,D}(x_*) \geq Lr = \frac{1}{2}|\mathcal{B}|.$$

The orbit-reflection block suppresses selected old fibres but necessarily creates a new linear-scale peak elsewhere.

4.4 The remaining obstacles

The affine/two-dissociative route reduces a possible improvement to a concrete structural selection problem, but the following points remain unresolved.

- (i) **Approximate periods are not automatic.** Two-dissociation models independence. It controls collisions of Fourier words but does not produce a frequency q with $\Re \hat{\mu}(q)$ close to 1.

- (ii) **Relative dissociation is required.** The relevant measure is the defect measure ν_q , not Haar measure. Ordinary two-dissociation of D does not by itself force the Fourier avoidance assumptions in [theorem 4.1](#).
- (iii) **The first moment has a sign constraint.** A small second moment is insufficient unless the block also has nonpositive Gibbs mean, or a separate mechanism controls the linear term.
- (iv) **Escape peaks must be treated dynamically.** Both simple pairs and orbit-reflection blocks create new large peaks. An iteration must suppress the new peak without reviving the old ones.
- (v) **General entropy extraction may consume the gain.** The ρ^{-2} cost in large-spectrum dimension bounds can offset a one-power correlation gain. A successful proof likely needs a theorem specific to Gibbs densities generated by positive sine sums, rather than a generic Chang-type estimate.

5 Numerical experiments

The numerical calculations are exploratory only. Complete scripts are included with the source package.

5.1 Bessel–Gibbs greedy search

For a practical deterministic search, the exact Gibbs score of a candidate n is

$$\int e^{\beta \sin(nx)} d\mu_j(x).$$

The Fourier–Bessel expansion

$$e^{\beta \sin t} = I_0(\beta) + 2 \sum_{k \geq 1} I_k(\beta) \cos \left(kt - \frac{k\pi}{2} \right)$$

allows all candidate scores to be evaluated simultaneously from one FFT of the current Gibbs density.

One run used

$$\beta = 2, \quad n \leq 150000, \quad N \leq 120.$$

The first forty selected frequencies were

$$\begin{aligned} &1, 3, 7, 17, 25, 45, 71, 13, 97, 9, 123, 265, 2, 132, 231, 399, \\ &250, 746, 666, 303, 6, 31, 1066, 89, 4, 76, 589, 282, 1423, 1777, \\ &1170, 1919, 2674, 644, 2597, 428, 3269, 3916, 576, 955. \end{aligned}$$

5.2 Gibbs-spectrum diagnostics

For each prefix, the Gibbs measure was recomputed at the theoretical scale

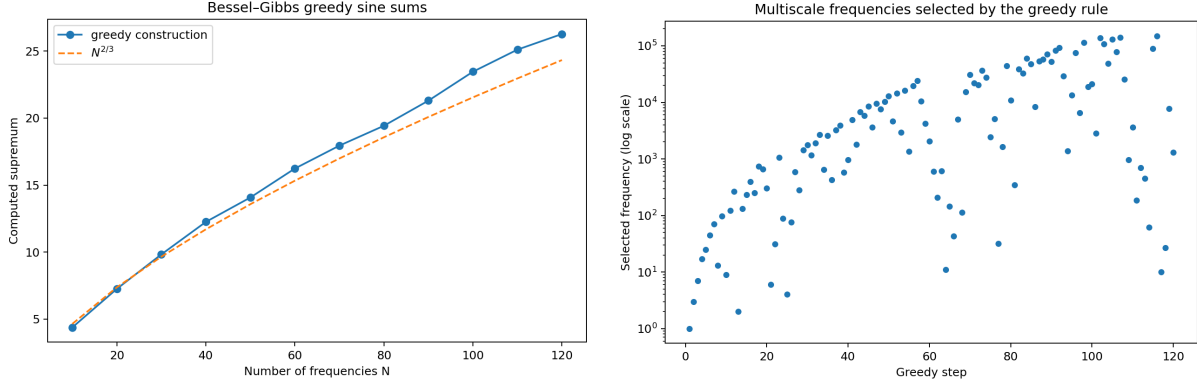
$$\beta = N^{-1/3},$$

and its Fourier coefficients were scanned for $1 \leq q \leq 150000$.

This does not disprove the affine-block strategy, but it shows that a strong near-period is not automatically present in this particular good $N^{2/3}$ -scale configuration.

Table 1: Computed prefix maxima for the Bessel–Gibbs greedy run.

N	computed maximum	$N^{2/3}$	ratio
20	7.2529	7.3681	0.9844
40	12.2512	11.6961	1.0475
60	16.2339	15.3262	1.0592
80	19.4252	18.5664	1.0463
100	23.4589	21.5443	1.0889
120	26.2731	24.3288	1.0799



(a) Computed maximum compared with $N^{2/3}$.

(b) Selected frequency scales.

Figure 1: The greedy output is multiscale rather than purely lacunary or confined to one multiplicative band.

5.3 A sample affine-block diagnostic

For the prefix $N = 120$, with $\beta = N^{-1/3}$, the code tested $q = 206$ and the two-dissociated set

$$D = \{1, 5, 25\}.$$

The base-5 digit argument shows that D is two-dissociated. Numerically,

$$1 - \mathfrak{R}\hat{\mu}(206) = 0.9661,$$

so $q = 206$ is not a near-period. The simple pair block had Gibbs mean 0.1918, second moment 3.0108, and log Gibbs multiplier 0.1004. The $L = 2$ orbit-reflection variant had a larger second moment and multiplier. The exact factorisations were verified to errors below 10^{-12} on the grid. This small calculation illustrates the main obstruction: two-dissociation controls word collisions, but without a strong affine near-period it does not create negative correlation.

6 Conclusions

The rigorous outcomes of this note are:

- (1) a closed deterministic proof of the known $N^{2/3}$ upper-bound scale, with frequencies below $\exp(O(N^{1/3}))$;

Table 2: Largest nonzero Gibbs Fourier coefficients on the scanned range.

N	q for max $ \widehat{\mu}(q) $	max $ \widehat{\mu}(q) $	q for max $\Re \widehat{\mu}(q)$	max $\Re \widehat{\mu}(q)$
20	303	0.1756	6	0.0659
40	3	0.1416	1	0.0677
60	10340	0.1243	1	0.0521
80	2	0.1186	2	0.0680
100	4	0.1093	2	0.0600
120	4	0.1130	2	0.0663

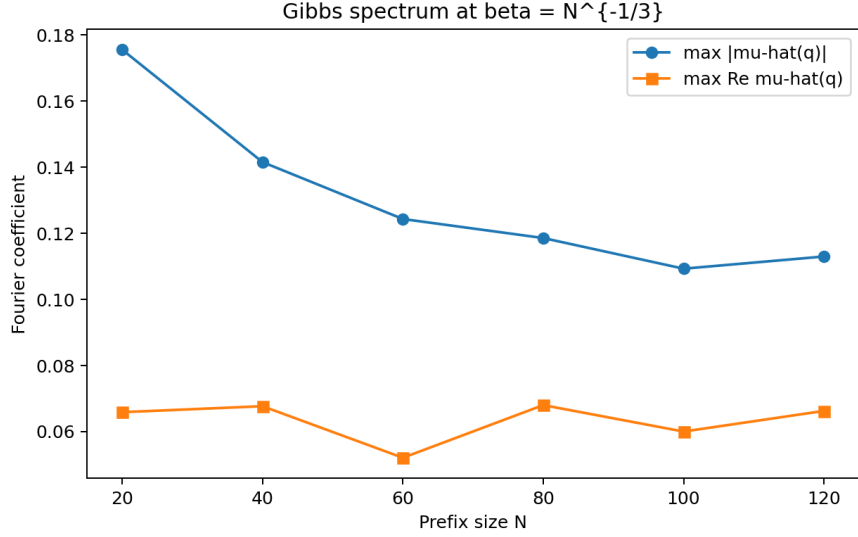


Figure 2: No strong approximate period is visible at the natural Gibbs scale in this run.

(2) the structural obstruction

$$\|S_A\|_\infty \geq \frac{\dim_2(A)^2}{4|A|};$$

(3) the entropy large-spectrum bound

$$\dim_2(\text{Spec}_\rho(f)) \leq \frac{4 \text{Ent}(f)}{\rho^2};$$

(4) exact affine-pair and orbit-reflection factorisations, together with a conditional block mechanism that would improve the exponent if a uniform sublinear-variance selection theorem could be proved.

The most focused remaining problem is the following.

Conjectural principle 6.1 (Adaptive affine block principle). *There exist $\eta, c, C > 0$ such that, for every Gibbs state arising from a suitably designed construction at parameter β , one can find $r \asymp \beta^{-1}$ fresh affine pairs below $e^{C/\beta}$ whose block G satisfies*

$$\int G \, d\mu \leq 0, \quad \int G^2 \, d\mu \leq Cr^{1-\eta}.$$

By [theorem 4.3](#), this principle would yield a strict improvement over $N^{2/3}$. The present work identifies the exact algebraic and Fourier conditions needed for such a block, but does not prove their uniform availability.

References

- [1] M. N. Kolountzakis, *A remark on perturbations of sine and cosine sums*, 1999, arXiv:math/9912149. <https://arxiv.org/abs/math/9912149>
- [2] B. Bedert, *Riesz products and the Lonely Runner Conjecture: A wider gap of loneliness*, 2025, arXiv:2511.16636. <https://arxiv.org/abs/2511.16636>